

Gagandeep Shivanna

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☎ +49 15560731320 📍 Berlin, Germany 🇮🇳 Nationality: Indian | 🎓 Visa: Student (Work permitted)

Profile

AI engineer with **2+ years** of production java backend experience, now focused on **LLM evaluation**, **RAG pipelines**, and shipping GenAI features that survive contact with real users. Master's thesis is an empirical study of verifier-guided LLM self-repair across three frontier models. Builds with backend discipline, sandboxed execution, eval harnesses, and **Infrastructure-as-Code**. Constant investment of time in end to end projects.

Work Experience

Sovrano AI

06/2026 – Present

AI Trainer Analyst

Remote

- Designed and applied **rubric-based evaluation** methodology for LLM output quality across accuracy, relevance, and safety dimensions, building **gold-standard reference sets** used to score model responses.
- Ran structured side-by-side **RLHF-style model comparisons** and **pairwise preference evaluation**, producing the labelled preference data used for **reward-model training**.
- Authored **task decomposition specs** at intent level for **multi-step agent workflows**, improving inter-annotator consistency across the evaluation pipeline.

SRH University of Berlin

09/2025 – 04/2026

Research Assistant

Berlin, Germany

- Built end-to-end **RAG pipelines** for document classification and information extraction, using **ChromaDB** for vector storage and **semantic retrieval** over domain document corpora.
- Tuned **chunking strategy**, **embedding** selection, and retrieval depth against a held-out evaluation set, improving extraction accuracy over a zero-shot baseline.
- Applied **prompt engineering** and **structured output constraints** to make extraction results machine-parseable and reproducible across model versions.

Vrize India Pvt Ltd

09/2022 – 07/2024

Backend Engineer

Bangalore, India

- Developed and maintained enterprise-level **Java/Spring Boot** applications (Zaxby's, Tonic), designing scalable **RESTful APIs** and backend **microservices** with database integration across **MySQL** and **MongoDB**.
- Architected modular backend services implementing **Dependency Injection** and **AOP** patterns to improve maintainability and reduce coupling across service layers.
- Led API debugging and unit/integration testing using **JUnit** and **Mockito**, identifying and resolving **50+ critical bugs** and enforcing code quality standards through structured peer reviews.
- Automated deployment pipelines using **Docker** and **Jenkins** within **CI/CD** workflows, reducing manual release overhead and improving deployment reliability.
- Contributed to **15+ Agile sprint cycles**, consistently meeting milestones and recognized by senior engineers for technical ownership and cross-functional collaboration.

Education and Training

SRH University of Berlin

07/2024 – Expected 09/2026

M.Sc. Big Data and Artificial Intelligence (120 ECTS) – CGPA: 1.4

Berlin, Germany

- **Master Thesis:** Built a failure taxonomy of **LLM-generated Infrastructure-as-Code**, running a three-stage verification pipeline (**Terraform**, **Checkov**, **tfsec**, **OPA/Rego**) via AWS with an automated **self-repair loop** across **542 runs** and three models. Analyzed **2,000+** logged errors using **survival analysis** and significance testing to identify which error categories resist verifier-guided repair, validated against the **IaC-Eval** benchmark.

- **Research Paper (Ohrid Conference)**: Built a stacking ensemble combining **LSTM** and **Random Forest** predictions in an **XGBoost** meta-learner to test whether Tesla supplier stock movements carry predictive signal for Tesla's own price, achieving $R^2 = 0.9558$ and a **14% RMSE reduction** over the single-model baseline.

BMS College of Engineering, Bangalore

B.E. Information Science and Engineering – CGPA: 8.17

08/2018 – 07/2022

Bangalore, India

Skills

Programming	Python, Java, JavaScript/TypeScript, SQL, C, C++
LLM & GenAI	RAG, LangChain, Claude API, OpenAI API, Ollama, Prompt Engineering, Agent Workflows
Vector & Retrieval	ChromaDB, Embeddings, Semantic Search, Chunking Strategy
LLM Evaluation	Eval Harness Design, RLHF Preference Data, Rubric Design, MLflow, Text-to-SQL Eval
AI/ML	PyTorch, TensorFlow, Scikit-learn, XGBoost, Keras, Transformers, BERT, LSTMs
Development	Spring Boot, FastAPI, RESTful APIs, Microservices, WebSockets, MySQL, MongoDB, Node.js
Cloud & DevOps	AWS, GCP, Azure, Terraform, Infrastructure-as-Code (IaC), Docker, Jenkins, CI/CD, GitHub Actions, Render, Vercel
Testing	JUnit, pytest, Checkov, tfsec, OPA/Rego, Cross-validation, Test Automation
Tools	Git, Jira, PowerBI, Tableau, VS Code, Confluence, Jupyter, Recharts

Projects

DataGrid: Production Text-to-SQL Analytics Platform

[Live Demo](#) | [GitHub](#)

FastAPI, React, DuckDB, Claude Sonnet/Haiku API, MLflow, SDV

- Deployed a **natural-language-to-SQL** platform where the model writes the query, **DuckDB** executes it sandboxed, and a second model streams the interpretation over **SSE**. Built a **text-to-SQL eval harness** measuring **execution accuracy** by result-set comparison rather than string match, logged to **MLflow**.

AsyncMeet (MANAN): AI Async Decision-Making Platform

Active Development

FastAPI, React/TypeScript, Claude Sonnet API, WebSockets

- A **SaaS platform** replacing live meetings, where a guided **LLM** chat generates its own **follow-up questions** to close gaps in participant input. Responses are synthesized into a structured decision summary on a **real-time WebSocket dashboard**.

AWS Account Onboarding Automation

[GitHub](#)

Terraform, GitHub Actions (OIDC), AWS, Python

- An **Infrastructure-as-Code** pipeline provisioning a full AWS account from a single YAML config, **IAM**, **S3**, and **VPC**, with **OIDC auth** and no static credentials. Its reference configs became the **Tier B benchmark set** for my thesis eval harness.

Certifications

- Google Cloud [View certifications](#)
- AWS Educate [View certifications](#)

Language Skills

Languages: English (C2), German (B1)